True or false? (unless otherwise specified.) Questions 1–15: 1 point for a correct answer, 0 otherwise.

Answer grid

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	B
True	☐	☒	☒	☒	☒	☒	☐	☒	☐	☐	☐	☒	☐	☒	☐	☒
False	☒	☐	☐	☐	☐	☐	☒	☐	☒	☒	☒	☐	☒	☐	☒	☐

1. **Degree of precision.** The degree of precision of the following quadrature rule is 2:

$$\int_{-1}^{1} f(x)\,\mathrm{d}x \approx \frac{1}{2}f(-1) + f(0) + \frac{1}{2}f(1)\,.$$

False. The rule is exact for 1 ($\frac{1}{2}+1+\frac{1}{2}=2$) and, by symmetry, for x. For x^2 it gives $\frac{1}{2}+\frac{1}{2}=1\neq\frac{2}{3}$, so the degree of precision is 1.

2. **Degree of precision.** The degree of precision of the following quadrature rule is 1:

$$\int_{0}^{1} f(x)\,\mathrm{d}x \approx f(0) + \frac{1}{2}f'(0)\,.$$

True. The rule is exact for 1 (it gives 1) and for x (it gives $0+\frac{1}{2}=\frac{1}{2}$). For x^2 it gives $0\neq\frac{1}{3}$.

3. **Degree of precision.** Let $x_1 < \cdots < x_n$ be distinct nodes in $[-1,1]$. If the quadrature rule

$$\int_{-1}^{1} f(x)\,\mathrm{d}x \approx \sum_{i=1}^{n} w_i f(x_i)$$

has degree of precision at least $n-1$, then its weights are given by

$$w_i = \int_{-1}^{1} \ell_i(x)\,\mathrm{d}x, \qquad \ell_i(x) = \prod_{\substack{j=1\\ j\neq i}}^{n} \frac{x-x_j}{x_i-x_j}, \qquad i=1,\dots,n.$$

True. The Lagrange polynomial ℓ_i belongs to $\mathcal{P}_{n-1}$, so the rule is exact for ℓ_i. Since $\ell_i(x_j)=\delta_{ij}$, this gives

$$\int_{-1}^{1} \ell_i(x)\,\mathrm{d}x = \sum_{j=1}^{n} w_j \ell_i(x_j) = w_i.$$

4. **Gaussian integration.** The following quadrature rule is the two-point Gauss–Legendre rule:

$$\int_{-1}^{1} f(x)\,\mathrm{d}x \approx f\left(-\frac{1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}\right). \tag{1}$$

True. The nodes of the two-point Gauss–Legendre rule are the roots of the Legendre polynomial $P_2(x) = \frac{1}{2}(3x^2-1)$, that is $\pm\frac{1}{\sqrt{3}}$. Exactness for 1 gives $w_1+w_2=2$, and exactness for x gives $w_1=w_2$, so $w_1=w_2=1$.

5. **Composite quadrature.** Suppose that $f\in C^\infty[a,b]$ and let $I_n[f]$ denote the approximation of the integral $I[f] \coloneqq \int_a^b f(x)\,\mathrm{d}x$ obtained with the composite two-point Gauss–Legendre rule with n cells, that is

$$I_n[f] = \frac{h}{2}\sum_{i=0}^{n-1}\left(f\left(m_i - \frac{h}{2\sqrt{3}}\right) + f\left(m_i + \frac{h}{2\sqrt{3}}\right)\right), \qquad h=\frac{b-a}{n}, \qquad m_i = a + \left(i+\frac{1}{2}\right)h\,.$$

Then there exists a constant $C\in(0,\infty)$ such that

$$\forall n \in \{1,2,\dots\}, \qquad \left|I[f]-I_n[f]\right| \leqslant \frac{C}{n^4}\,.$$

Hint: Compute or recall the degree of precision of the rule (1), and use the theorem relating the degree of precision of a quadrature rule to the order of convergence, with respect to h, of the associated composite rule.

True. The two-point Gauss–Legendre rule is exact for 1, x and x^3 by symmetry, and for x^2 since $\frac{1}{3}+\frac{1}{3}=\frac{2}{3}$; it is not exact for x^4 ($\frac{2}{9}\neq\frac{2}{5}$). Its degree of precision is $d=3$, so the composite rule converges with order $d+1=4$ with respect to h, that is, the error is $O(h^4)=O(n^{-4})$.

6. **Composite quadrature.** With the notation of the previous item, it holds that $I_n[f] = I[f]$ for all $n \geqslant 1$ whenever f is a polynomial of degree at most 3.

   True. On each cell, the rule is the two-point Gauss–Legendre rule mapped to that cell, which is exact for polynomials of degree at most 3. Summing over the cells, the composite rule is exact for these polynomials too.

7. **Composite quadrature.** Assume that the error of a composite quadrature rule with n cells behaves like Cn^{-2} for large n, and that the error obtained with $n = 100$ cells is approximately 10^{-2}. Then the error obtained with $n = 1000$ cells is approximately 10^{-6}.

   False. Multiplying n by 10 divides the error by $10^2 = 100$, so the error is approximately 10^{-4}. (Here $C \approx 10^{-2} \times 100^2 = 100$; the value $10^{-6} = 1000^{-2}$ would correspond to $C = 1$.)

8. **Gaussian integration.** The one-point Gauss–Legendre rule on $[-1, 1]$ is the midpoint rule

$$\int_{-1}^{1} f(x)\,\mathrm{d}x \approx 2f(0)\,.$$

   True. The one-point rule must be exact for 1 and x: this gives $w_1 = 2$ and $w_1 x_1 = 0$, so $x_1 = 0$. (Equivalently, $x_1 = 0$ is the root of the Legendre polynomial $P_1(x) = x$.)

9. **Gaussian integration.** The nodes of the n-point Gauss–Legendre rule on $[-1, 1]$ are equally spaced.

   False. The nodes are the roots of the Legendre polynomial of degree n, which are not equally spaced in general. For instance, for $n = 3$ the nodes are 0 and $\pm\sqrt{3/5}$.

10. **Gaussian integration.** As for closed Newton–Cotes rules, the n-point Gauss–Legendre rule has some negative weights when n is large enough.

    False. All the weights of the Gauss–Legendre rule are positive, for every n. Indeed, the Lagrange polynomial ℓ_i associated with the nodes satisfies $\ell_i^2 \in \mathcal{P}_{2n-2}$, which is integrated exactly, so

$$0 < \int_{-1}^{1} \ell_i(x)^2\,\mathrm{d}x = \sum_{j=1}^{n} w_j \ell_i(x_j)^2 = w_i.$$

11. **Gaussian integration.** Let $x_1, \ldots, x_n$ and $w_1, \ldots, w_n$ denote the nodes and weights of the n-point Gauss–Legendre rule on $[-1, 1]$. Then the n-point Gauss–Legendre rule on a general interval $[a, b]$ is given by

$$\int_a^b f(x)\,\mathrm{d}x \approx \sum_{i=1}^{n} w_i f\left(a + \frac{b-a}{2}(x_i + 1)\right).$$

    False. The nodes are correctly mapped, but the weights must be multiplied by $\frac{b-a}{2}$, the Jacobian of the affine change of variables. For instance, with $f = 1$ the formula gives $\sum_i w_i = 2$ instead of $b - a$.

12. **Gaussian integration.** There exist a node $x_2 \in (-1, 1]$ and weights w_1, w_2 such that the rule

$$\int_{-1}^{1} u(x)\,\mathrm{d}x \approx w_1 u(-1) + w_2 u(x_2)$$

    has a degree of precision equal to 2.

    True. Exactness for 1, x and x^2 gives the equations

$$w_1 + w_2 = 2, \qquad -w_1 + w_2 x_2 = 0, \qquad w_1 + w_2 x_2^2 = \frac{2}{3}.$$

    The second equation gives $w_1 = w_2 x_2$, and then the first gives $w_2 = \frac{2}{1+x_2}$. Substituting in the third equation, $w_2 x_2 (1 + x_2) = 2x_2 = \frac{2}{3}$, so $x_2 = \frac{1}{3}$, $w_2 = \frac{3}{2}$ and $w_1 = \frac{1}{2}$. The rule is not exact for x^3, since it gives

$$-\frac{1}{2} + \frac{3}{2} \cdot \frac{1}{27} = -\frac{4}{9} \neq 0 = \int_{-1}^{1} x^3\,\mathrm{d}x,$$

    so its degree of precision is exactly 2. (This is the two-point Gauss–Radau rule.)

13. **Richardson extrapolation.** Let $T(h)$ be the composite trapezoidal approximation of $I := \int_a^b f(x)\,\mathrm{d}x$ with step size h, and assume that $T(h) = I + c_2h^2 + c_4h^4 + O(h^6)$ with $c_2 \neq 0$. Then the approximation

$$R(h) := 2T(h) - T(2h)$$

satisfies

$$\big|R(h) - I\big| = O(h^4)\,.$$

False. We have

$$R(h) = 2T(h) - T(2h) = I + c_2(2-4)h^2 + O(h^4) = I - 2c_2h^2 + O(h^4),$$

which is still second order. The correct combination is $\frac{4T(h)-T(2h)}{3}$.

14. **Richardson extrapolation.** With the notation and assumptions of question 13, it holds that

$$I - T(h) = \frac{T(h) - T(2h)}{3} + O(h^4)\,.$$

In other words, the error of $T(h)$ can be estimated from $T(h)$ and $T(2h)$.

True. We have

$$T(h) - T(2h) = c_2(1-4)h^2 + O(h^4) = -3c_2h^2 + O(h^4), \qquad I - T(h) = -c_2h^2 + O(h^4).$$

15. **Julia.** The following code implements the composite midpoint rule with n points:

```
function M(f, a, b, n)
    h = (b - a) / n
    x = LinRange(a + h/2, b - h/2, n + 1)
    return h * sum(f, x)
end
```

False. `LinRange(a + h/2, b - h/2, n + 1)` contains $n+1$ points, whereas there are only n midpoints. The correct code uses `LinRange(a + h/2, b - h/2, n)`.

16. **Bonus.** Let $f \in C^\infty(\mathbb{R})$ and $x \in \mathbb{R}$, and let $D(h)$ denote the central finite difference approximation of $f'(x)$:

$$D(h) = \frac{f(x+h) - f(x-h)}{2h}\,.$$

Then the approximation

$$\frac{4D(h) - D(2h)}{3}$$

satisfies

$$\left|\frac{4D(h) - D(2h)}{3} - f'(x)\right| = O(h^4) \qquad \text{as } h \to 0\,.$$

Justify briefly.

True. By Taylor's formula,

$$f(x \pm h) = f(x) \pm f'(x)h + \frac{f''(x)}{2}h^2 \pm \frac{f'''(x)}{6}h^3 + \frac{f^{(4)}(x)}{24}h^4 \pm \frac{f^{(5)}(x)}{120}h^5 + O(h^6),$$

so the even terms cancel in $D(h)$ and

$$D(h) = f'(x) + \frac{f'''(x)}{6}h^2 + O(h^4), \qquad D(2h) = f'(x) + \frac{4f'''(x)}{6}h^2 + O(h^4).$$

Therefore $4D(h) - D(2h) = 3f'(x) + O(h^4)$: the h^2 term cancels.